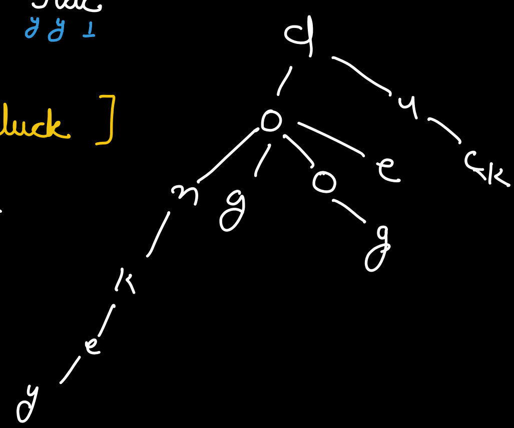
Q Given an array of strings. Return an array containing the shortest unique prefix for every string in the given array.

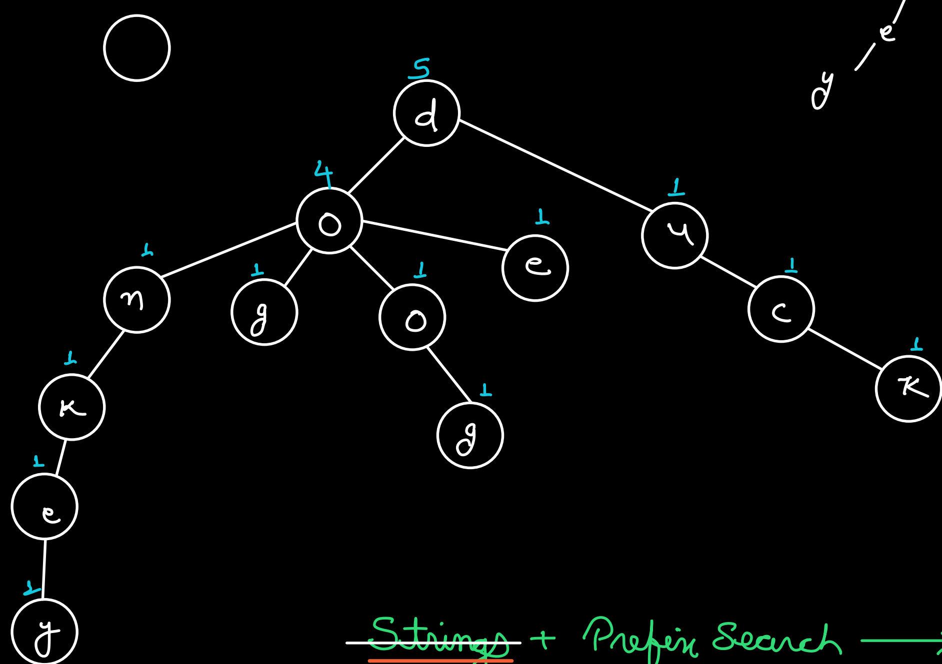
A : [cat, dog, rat, tiger, racoon]

c , d , rat , t , rac

A : [dog, doog, doe, donkey, duck]

dog , doo , doe , don , du





~~Strings~~ + Prefix Search ⟶ Tries

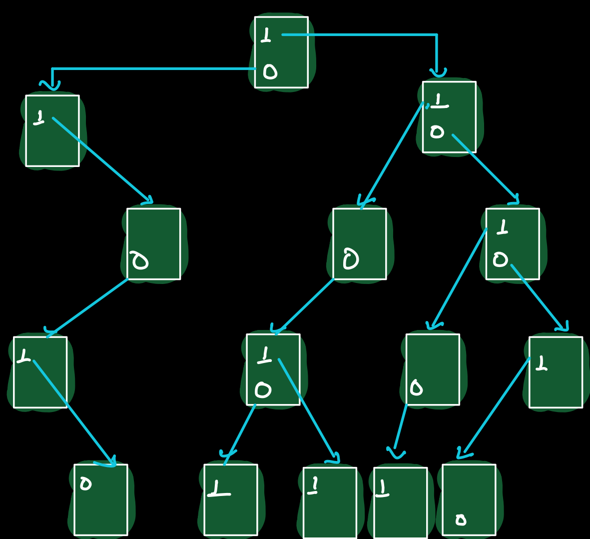
Tries are trees

Q Given a binary matrix find the no. distinct rows.

0	1	0	0	1	0
1	0	1	0	1	0
2	1	1	0	1	1
3	1	1	0	0	1
4	1	1	0	1	1
5	1	0	0	1	0
6	1	0	1	0	1
7	0	0	1	1	0

$\longrightarrow 6$

$N \times M$



NK

insert(row)

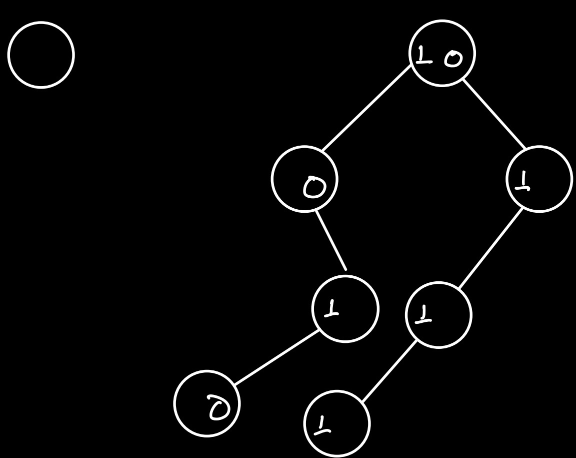
TC : O(M)

search(row)

TC : O(M)

TC : O(NM)

SC : O(NM)



0 1 1 0

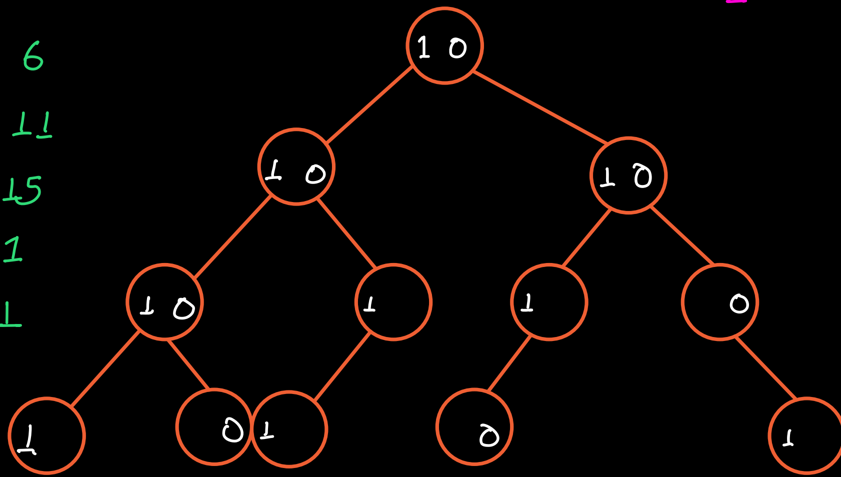
1 0 1 0

Child of 1 → left

Child of 0 → right

11	1011	6
6	0110	11
1	0001	15
12	1100	1
15	1111	1

0001



Q Given an array of positive integers. Return the max XOR of any pair.

0	1	2	3
4	3	2	7

$4 \wedge 3 \longrightarrow 7$

0	1	2	3
8	9	7	1

$8 \wedge 7 \longrightarrow 15$

<u>Brute force</u>

TC: $O(N^2)$

SC: $O(1)$

8	:	1	0	0	0
9	:	1	0	0	1
7	:	0	1	1	1
1	:	0	0	0	1

1 0

0 1

1 1

1, 4, 8, 15

	22	61	38	27	21	34	42	37	43
	5	4	3	2	1	0			
22	0	1	0	1	1	0	43		
61	1	1	1	1	0	1	22		
38	1	0	0	1	1	0	27		
27	0	1	1	0	1	1	38		
21	0	1	0	1	0	1	42		
34	1	0	0	0	1	0	27		
42	1	0	1	0	1	0	21		
37	1	0	0	1	0	1	27		
43	1	0	1	0	1	1	21		

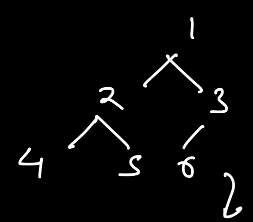
TC: $O(\underbrace{\text{Size of data type}} \times N)$

$\downarrow$

$O(N \log \text{Max})$

Revision

Storing BT in array



1	2	3	4	5	6

parent ← i → left child, right